



**unilog**

**h2s**  
HACK2SKILL

# UniHack

AI-Powered Product Intelligence for  
**Industrial Commerce**

## Team Details

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# An evidence-gated compiler for product data.

AXIOM turns **six sparse input fields** into a governed, reviewable, delivery-ready catalog record — **without guessing**.

## 01 · PROBLEM

### Sparse item masters. Rigid downstream contracts.

- Only a part number, conflicting brand fields, a supplier/manufacturer label.
- Commerce still demands structured specs, taxonomy, descriptions, URLs, and the rigid 252-column contract.

## 02 · SOLUTION

### Retrieve. Classify. Govern.

- Search manufacturer evidence from a part number alone.
- Lock the right schema or abstain rather than guess it.
- Extract, normalize, validate; score per field.

## 03 · OUTPUT

### Auditable delivery, not just a fuller row.

- versioned record · human-review queue · enrichment certificate.
- Exact 252-column UniLog file with per-cell provenance sidecar.

# How does AXIOM turn six sparse fields into rich product intelligence?

*A staged pipeline — retrieve first, classify before extraction, decide per field — so published facts are tied to evidence, not to language-model confidence.*

## 01 • IDENTITY

### Ingest minimal input.

Accept a part number and manufacturer — plus optional description, brand, URL, PDF. Batch mode reads CSV, TSV, XLSX item masters.

## 02 • EVIDENCE

### Acquire — four layers, all model-free.

Content-hashed library → manufacturer site/sitemap → Brave or keyless DDG → bounded Playwright 1.62. Search snippets discarded — only source bytes count.

## 03 • PARSE + HASH

### Lock the source before reasoning.

Text, HTML, tables, optional PDF → common representation. Provenance and content hash recorded. SKU verified exactly or via separator-normalized match before extraction.

## 04 • CLASSIFY

### Select the right schema — or abstain.

Rank product-class candidates from description and document. If no class is defensible, return an identity-only record rather than inventing one.

## 05 • EXTRACT

### Schema-bound, tiered cascade.

Deterministic extraction + AWS Bedrock cascade (ZAI GLM-4.7 flash → 4.7 → 5). Cheapest tier runs first; escalates only on validator rejection — cost follows difficulty.

## 06 • NORMALIZE + VALIDATE

### Make data consistent and rule-checked.

Canonicalize quantities, units, ranges, enums, identifiers. Apply type, range, unit, required-field, and cross-field validation — input is a candidate hint, never its own proof.

# The missing trust layer between source documents and the PIM.

Positioning hypothesis — not an external benchmark. Generic AI excels at coverage; AXIOM couples coverage with field-level verifiability.

| APPROACH             | AUTOMATION | FIELD-LEVEL EVIDENCE  | KNOWS WHEN TO STOP          |
|----------------------|------------|-----------------------|-----------------------------|
| Manual research      | Low        | Variable              | Yes                         |
| Rules / RPA          | Medium     | Variable              | Limited                     |
| Generic LLM autofill | High       | Often weak            | Often no                    |
| AXIOM                | High       | Built into each value | Abstain · review · withhold |

Every field carries its own evidence, validation, confidence, and publication state — *and the system knows when not to guess.*

# How does AXIOM ensure accuracy and trust for catalog data?

Layered controls around one rule: *a value is published only when evidence, validation, and policy all clear it* — never on a model's self-reported confidence alone.

10 • LAYERED CONTROLS, APPLIED PER FIELD

|                                                                                                                                                                                        |                                                                                                                                                                                                         |                                                                                                                                                                                         |                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>01</div> <div>Evidence contract.</div> <div>Each value carries at least one evidence span: source document, exact quote, content hash, page or location.</div>                    | <div>02</div> <div>Quote verification.</div> <div>The system records whether the cited text can be matched back to the source — verified or unverified, never silently mixed.</div>                     | <div>03</div> <div>Input is not proof.</div> <div>The customer's own item-master description is a candidate and retrieval hint — never independent evidence for publication.</div>      | <div>04</div> <div>Classification abstention.</div> <div>If no product class is defensible, AXIOM returns an identity-only record rather than hallucinating a class for extraction.</div> |
| <div>05</div> <div>Rule-based validation.</div> <div>Type, range, unit, enum, identifier, constraint, cross-field consistency — all checked after normalization, before scoring.</div> | <div>06</div> <div>Calibrated risk policy.</div> <div>Evidence + validation features feed a policy with a stated error bound. Cold start defaults to review until automation has earned evidence.</div> | <div>07</div> <div>Inference restrictions.</div> <div>Part-number patterns, family inference, and statistical defaults cannot auto-publish silently — they demand human approval.</div> | <div>08</div> <div>Multi-source conflict handling.</div> <div>Sources are compared; conflicts escalated or reviewed. Sourced values are not averaged into false certainty.</div>          |

|                                                                                                                                                      |                                                                                                                                                                           |                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>09 • HUMAN REVIEW</div> <div>Reviewer accepts, corrects, or rejects — decision and timestamp recorded, model and schema versions stamped.</div> | <div>10 • FINAL GATE</div> <div>High confidence alone is insufficient. Failed validation, missing evidence, or self-declared input keeps the value out of delivery.</div> | <div>MEASURED PROTOTYPE BACKTEST</div> <div> <div>15 products</div> <div>312 comparisons</div> <div>99.1 PRECISION</div> <div>100 CITATION COVERAGE</div> <div>0 HALLUCINATED OUTPUTS</div> </div> |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Prototype backtest on a small labeled corpus — not a production guarantee. Honest qualification, on every metric, every claim.

# What makes AXIOM scalable for enterprise product catalogs?

*Software boundaries designed for scale — modular stages, config-driven schemas, content-hashed sources — with a measured coverage number and an honest production-hardening roadmap.*

## 6 • DIMENSIONS OF SCALABILITY

### 01 • LARGE CATALOGS

#### Batch + single, one pipeline.

Stages are separable. Retrieval, extraction, validation, export can become independent workers — a production deployment swaps the synchronous path for durable queues, idempotent jobs, retries, and horizontally scalable workers.

### 02 • NEW MANUFACTURERS

#### Modular providers, no code touch.

Aliases and domains added through config; source-authority rules are pluggable. Measured on 1,000 real item-master rows: 68.1% resolve to a declared manufacturer domain, 12.1% to the stored library.

### 03 • NEW PRODUCT CLASSES

#### Registry-driven, not hard-coded.

Classes, attributes, labels, slot bindings, constraints, and mappings live in registry and config. New categories require schemas and evaluation data — not a new application.

### 04 • NEW DOCUMENTS

#### Ingestion is separable.

CSV / TSV / XLSX item-master parsing is independent from text, HTML, table, and PDF processing. OCR / vision can be added upstream while downstream governance stays intact.

### 05 • CONTINUOUS UPDATES

#### Versioned, never overwritten.

Sources are content-hashed; values superseded, not overwritten. Model, prompt, and schema versions recorded. Changed sources can be reprocessed and audited as diffs.

### 06 • COST CONTROL

#### Cost follows difficulty, not guess.

Deterministic stages cut unnecessary calls, classification abstention stops extraction early, the tiered cascade reserves expensive models for true ambiguity, and unchanged artifacts are cached by hash.

### MEASURED • 1,000 REAL ITEM-MASTER ROWS

**80.2%** reachable **without any search API**

68.1% resolve to a declared manufacturer domain • 12.1% to the stored document library • 10.3% need a search or one config entry • 9.5% name only a distributor — a data-quality issue upstream that no search provider can fix.

#### HONEST GAP

*Modular batch processing exists. High-volume distributed throughput, production SLOs, and secure multi-tenancy are honest hardening goals — not completed benchmarks.*

# A typed, testable, cloud-ready stack.

Hybrid by design — deterministic where it can be, AI-assisted where it adds value, human-reviewed where trust is unresolved.

## CORE

Python 3.11+  
Pydantic 2.13  
PyYAML

Typed values, schemas, rules, provenance.

## API

FastAPI 0.141  
Uvicorn 0.34

Enrichment, batch delivery, review endpoints.

## AI · MODELS

AWS Bedrock Converse  
ZAI GLM-4.7 flash → 4.7 → 5  
Qwen3-VL (vision)

Tiered cascade, used by **2 of 9 stages**.

## RETRIEVAL

Brave Search · DDG fallback  
Playwright 1.62 · robots  
URL, SSRF, byte policy

URL-only discovery + bounded JS rendering.

## DATA · DOCS

pdfplumber · openpyxl  
CSV · HTML · TSV parsers  
Content-addressed artifacts

Immutable provenance for every claim.

## CONSOLE

Next.js 16 · React 19  
TypeScript 7 · Tailwind 4

Review, certificate, pipeline, delivery UI.

## QUALITY

pytest · Vitest  
Ruff  
Evaluation + regression harnesses

Backend, UI, lint, behaviour gates.

## DELIVERY · INFRA

Docker · Compose  
GitHub Actions  
AWS CDK definitions

Reproducible packaging, CI, infrastructure as code.

## BOUNDARY

2 / 9

Stages call a model. *The rest is deterministic.*

Extraction always does. Classification only on a near-tie. Retrieval, parsing, normalization, validation, scoring, certification, and delivery never do. Enforced by a CI test, not a slide.

# From trusted prototype to production catalog infrastructure.

PRINCIPLE

*Trust before coverage. Three horizons, measurable gates.*

## HORIZON 1

### Harden & measure.

0 — 3 MONTHS

- Authentication, authorization, tenant isolation, audit logs.
- Durable async jobs, retries, idempotency, controlled egress.
- Larger independently labeled, class-stratified evaluations.
- Validate container image and live Brave provider.

## HORIZON 2

### Integrate & scale.

3 — 6 MONTHS

- PIM/ERP and supplier-feed connectors.
- Object storage, database state, tracing, metrics, alerts, SLOs.
- Reviewer productivity analytics and active calibration.
- More classes, more manufacturers, more delivery channels.

## HORIZON 3

### Governed intelligence.

6 — 12 MONTHS

- OCR / vision for scans and image-heavy documents.
- Multilingual extraction and localization policies.
- Change monitoring and evaluated equivalence workflows.
- Versioned fine-tuning only with approved data, holdouts, audit, rollback.

## PILOT SUCCESS METRICS — MEASUREMENT, NOT ASSERTION

• Precision & citation coverage *by class*   • Auto-accept · review · reject · withhold *rates*   • Reviewer minutes & correction *per SKU*   • Time-to-publish & cost per *publishable field*   • Source freshness & update *latency*

Links :

GitHub Public Repository : <https://github.com/varshith21-codes/Unilog>

Working Prototype Link : <https://axiom.vectoredu.online>

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# UniHack

AI-Powered Product Intelligence for  
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# Thank You

